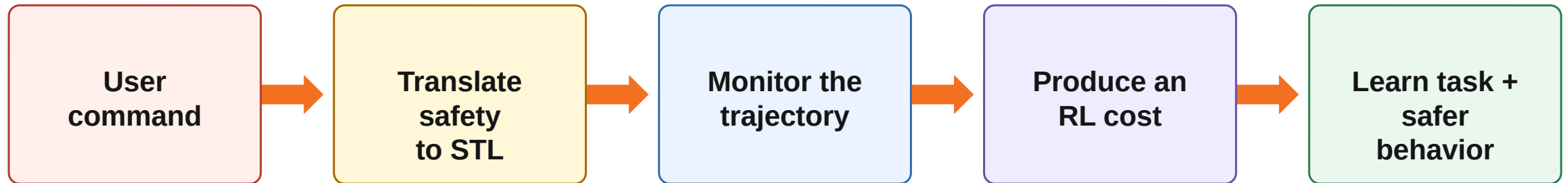


The Complete Goal: Language to Safer RL

1

"Reach the goal. If you get too close to a hazard, return to a safe distance within the allowed time."



"Too close -> recover in time" becomes a rule, a trajectory check, a safety cost, and finally a training signal.

Why the Research Is Split into Three Stages

2

The full chain contains three independent questions:

1. STL execution

Can a correct STL rule be monitored and used by RL?

2. Language translation

Can natural language be translated into STL accurately?

3. Broader setting

What changes with vague language, complex environments, and real robots?

If one end-to-end experiment fails, the cause is ambiguous:

Translation
was wrong

Grounding
was wrong

Monitor
was wrong

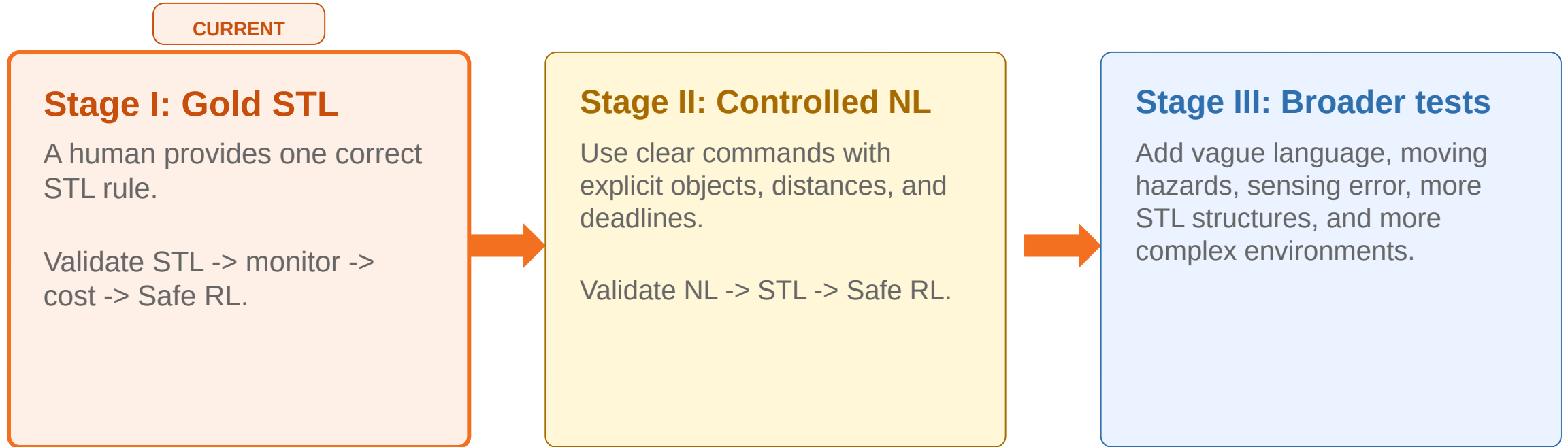
Cost never
reached RL

RL did not
learn

Separating the stages makes each failure interpretable.

Each Stage Answers One Question

3



Current work is Stage I: language translation is intentionally excluded.

Stage I Tests the Downstream Chain First

4

Can one correct STL rule monitor trajectories, produce a safety cost, and help Safe RL reduce temporal violations while still reaching the goal?



The Benchmark Fixes the Environment and Signal

5

SafetyPointGoal1-v0

A simple 2D navigation task

The environment contains

- one 2D Point agent
- one goal
- several static hazards
- task reward for reaching the goal
- native Safety-Gymnasium hazard cost

Our safety signal

**d_t = distance from the agent
to the nearest hazard center**

At every step, the monitor receives one distance value from the environment.

This fixes what is measured before we ask whether RL can learn from it.

One Rule Defines the Recovery Task

6

When the agent first enters the warning zone, it must return to the safe distance before the deadline.

Fixed parameters

warning distance = 0.45
safe distance = 0.55
deadline = 79 steps

After the first $d_t < 0.45$, the agent must achieve $d_t \geq 0.55$ within 79 steps.

Two distances create a recovery buffer

$d_t < 0.45$

Warning begins

$0.45 \leq d_t < 0.55$

Recovery is still pending

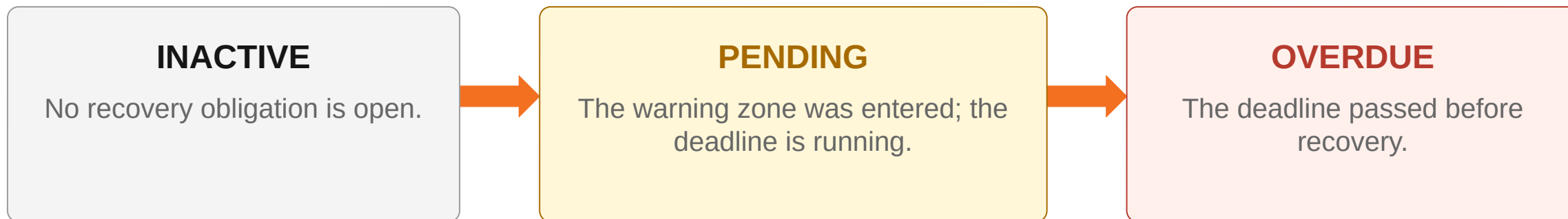
$d_t \geq 0.55$

Recovery is complete

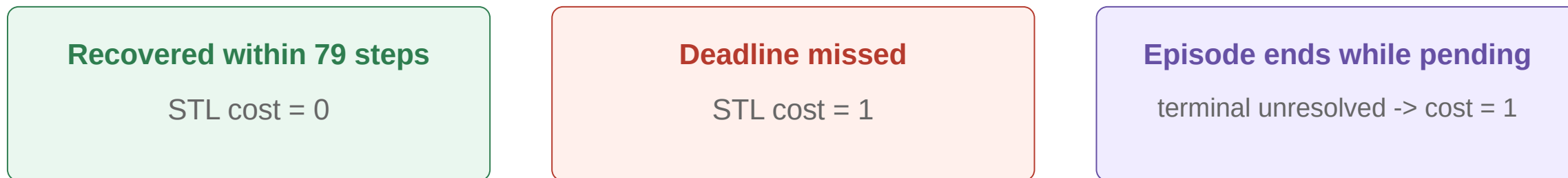
The Monitor Turns Recovery into an RL Cost

7

The monitor reads d_t at every step and keeps one recovery state:



The state becomes a cost only when the outcome is known:



This numerical cost is the signal that Safe RL can optimize.

Rule and Monitor Checks Are Complete

8

COMPLETED

Three independent checks agree on the declared cases:

Online monitor

Runs causally during the episode.

Direct offline oracle

Re-evaluates the saved trajectory independently.

RTAMT reference

Provides an external STL semantics check.

Why $K = 79$

95th percentile = 63 steps

$63 \times 1.25 = 78.75$

$\text{ceil}(78.75) = 79 \text{ steps}$

What the completed work proves

- 79 steps is evidence-based and generally achievable here.
- Environment, distance signal, monitor, cost, visualization, and logs run together.

It does not yet prove that a trained Safe RL policy will reduce violations.

Next: Build the OmniSafe Wrapper

9

The wrapper connects the tested STL cost to Safe RL without changing the benchmark task.



After the wrapper passes, train three matched agents:

TASK ONLY

NATIVE HAZARD COST

STL COST

Compare goal-reaching performance and temporal safety violations.